

Synthetic Data in the context of AI for Decarbonisation

A report by the AI for Decarbonisation Virtual Centre of Excellence
(ADViCE)

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The AI for Decarbonisation Virtual Centre of Excellence (ADViCE)

The ADViCE programme exists to bring together an ecosystem of AI companies, industrial adopters, policy makers and investors to support, promote, and co-ordinate the adoption and diffusion of AI decarbonisation applications. It is delivered by a consortium of partners – Digital Catapult, Energy Systems Catapult, and the Alan Turing Institute.

ADViCE takes a systems-led approach to accelerating net zero through AI. This means recognising that decarbonisation is a complex, interconnected challenge spanning technologies, sectors, infrastructure, policy, and behaviour. It recognises that whilst AI may increase efficiencies and improve processes to reduce emissions, the usage of AI itself also creates emissions. ADViCE therefore considers the opportunities and challenges of AI for decarbonisation as an integrated whole, prioritising based on system needs to support meaningful change at scale.

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Introduction

Artificial intelligence (AI) as a collection of technologies, has seen impressive and potentially transformative advancements in its capabilities in recent years. Whilst these advancements may create new ways of working and implications across our economy, there are still serious risks and limitations involved in how AI is trained, deployed, and utilised across society.

One inextricable issue in the efficacy of these technologies within the current paradigm is the availability of high-quality, representative, and well-labelled datasets. These are essential to train models that perform reliably, generalise well to new situations, and gain the trust of end-users. Without sufficiently complete and available data, AI systems risk being inaccurate, biased, or unsafe which can have serious consequences in critical sectors such as energy, the built environment, agriculture, transport, and manufacturing.

Yet in many domains, particularly those that are industrial, safety-critical, or carbon-intensive, obtaining the right data is challenging. Barriers include scarcity of examples of rare but important events, regulatory restrictions on sensitive data, high costs of collection and annotation, and fragmentation across siloed systems and organisations. These constraints can slow AI innovation, limit scalability, and prevent organisations from fully exploring the potential of AI solutions.

Why use synthetic data?

Synthetic data – novel, fabricated data generated by humans or machines to imitate a genuine dataset¹ – offers a potential avenue to address AI- and data-related challenges. In theory, it can replicate the statistical properties of real datasets while allowing greater flexibility, scalability, and privacy control. However, it is also important to note that whilst synthetic data may prove critical in unlocking innovation, the techniques used in building synthetic datasets come with their own risks and potential downstream negative consequences.

We have outlined below some of the major and consistent use cases for synthetic data. It is always important in any AI project to weigh the various pros and cons of these techniques from both a technical appropriateness and risk management perspective.

¹ https://www.turing.ac.uk/sites/default/files/2024-01/exploring_responsible_applications_report_november_2023_-_final_report.pdf

1. **Enabling easier data sharing and storage, as synthetic data is not subject to the same regulatory, privacy and protection requirements as some types of authentic data.**²

Synthetic data does not always preserve privacy by default,³ and in some cases the more a synthetic data model explicitly protects for privacy, the lower its functional utility.⁴

2. **Synthetically generated data is potentially more scalable and can augment data to compensate for data scarcity or missing data points.** Once a synthetic model is built, it can provide large volumes of training data and even generate novel examples.⁵

These novel examples and large volumes of data do not represent real world changes or 'ground truths', which could lead to inaccurate or unhelpful outputs in certain use cases.⁶

Incentivising the creation of large synthetic datasets increases the environmental impact of training and developing AI models.

3. **Using synthetic data also allows for lower barriers to entry to innovate and to test prototypes and new solutions.**⁷ High quality 'real' datasets are often proprietary or have various access requirements that effectively gatekeep assets. Open access synthetic datasets can allow for more researchers to benefit from the potential insights gained from data-intensive techniques.

Synthetic data generation methods that rely on a real dataset that is then 'augmented', or provides the initial information for the algorithmic creation of a new dataset, could still constitute an initial barrier to entry.

Balancing synthetic and real-world data

While synthetic data can address many challenges, it is not a universal substitute. Models trained solely on synthetic data have shown to produce undesirable outputs, and fail to capture unanticipated real-world variability, and in some instances validation of data

² <https://arxiv.org/pdf/2205.03257>

³ https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/05/oecd-digital-economy-outlook-2024-volume-1_d30a04c9/a1689dc5-en.pdf

⁴ <https://arxiv.org/pdf/2011.07018>

⁵ https://www.turing.ac.uk/sites/default/files/2024-01/exploring_responsible_applications_report_november_2023_-_final_report.pdf

⁶ Ibid.

⁷ Ibid.

remains critical.⁸ Approaches that also combine synthetic and real datasets can be considered but bring their own set of risks and opportunities in improving model performance, resilience, and trustworthiness.

The increase in use of synthetic data as a technique creates a substantial and hard to quantify environmental cost, due to the sheer volume of synthetic data that is required to train certain kinds of models (e.g., LLMs).^{9,10}

The potential viability of synthetic data may also encourage AI developers to continue to favour data-intensive workloads and models which would compound the environmental impact of AI, rather than potentially exploring and developing more compute efficient models trained on lower volumes of higher quality data.

This report aims to review the synthetic data landscape in the UK through the lens of AI and decarbonisation. The report includes:

- **An overview of synthetic data generation methods** and how these relate to performance criteria and responsible innovation considerations.
- **UK-based case studies** that have utilised synthetic data generation techniques for decarbonisation applications.
- **Opportunities and recommendations** for future initiatives around synthetic data in the UK's decarbonisation and AI innovation ecosystem.

⁸ <https://www.nature.com/articles/d41586-024-02420-7>

⁹ <https://blog.lboro.ac.uk/vice-chancellor/2024/10/07/everyone-must-understand-the-environmental-costs-of-ai/>

¹⁰ <https://alan-turing-institute.github.io/ADViCE/blog/How-do-we-balance-electricity-consumption-of-ai/>

Synthetic data generation - criteria and methods

Key criteria to consider when assessing different generation methods

The use of synthetic data seems to be a promising approach especially in fields such as healthcare, energy, finance etc., but there is still no consensus on there being a standardised approach for systematically evaluating the privacy or utility of synthetic data.¹¹ However, the methods that are used for synthetic data generation can be evaluated on seven commonly used criteria that assess different aspects of quality, appropriateness, and applicability. These criteria are **correctness, diversity, privacy, utility, fidelity, scalability, and compliance**.¹²

Correctness measures how accurately synthetic data replicates the statistical properties and patterns of real data. This includes maintaining proper distributions, correlations, and relationships between variables without introducing statistical artifacts or biases. Statistical artefacts are unintended distortions or spurious patterns that arise not from the true underlying data, but from AI model limitations that cause unrealistic extreme values or outliers introduced in the generated data. Biases are systematic deviations from the real data distribution. They occur, for example, in case real-world data has balanced classes, but the synthetic data generation process can create a class imbalance, excluding edge cases because the model focuses primarily on typical samples.¹³

Fidelity quantifies how closely synthetic data resembles real data in terms of visual, statistical, and structural characteristics. It indicates that synthetic data “looks and feels” like the real data, both quantitatively and qualitatively. High fidelity ensures that synthetic data is realistic and maintains the essential features that make it a viable substitute for real data¹⁴. Fidelity is therefore largely qualitative and perceptual, emphasising human interpretability and realism rather than strict statistical accuracy. For example, synthetic household energy consumption curves should display smooth, natural usage patterns like real homes, rather than unnatural spikes or irregular shapes, otherwise, the data would be considered low fidelity despite having correct statistical properties.¹⁵

Diversity evaluates the range and variety of synthetic samples generated, ensuring the method can produce sufficiently varied data points that cover the full spectrum of the

¹¹ <https://doi.org/10.1038/s41746-024-01359-3>

¹² <https://cy.ons.gov.uk/methodology/methodologicalpublications/generalmethodology/onsworkingpaperseries/onsmethodologyworkingpaperseriesnumber16syntheticdatapilot#synthetic-dataset-spectrum>

¹³ <https://pmc.ncbi.nlm.nih.gov/articles/PMC11958975/#B55>

¹⁴ <https://www.sciencedirect.com/science/article/pii/S2589004222016030>

¹⁵ <https://arxiv.org/pdf/2205.03257>

original data distribution while avoiding mode collapse or repetitive patterns.¹⁶ Mode collapse occurs when a generative model (e.g., GAN) produces synthetic data that represents only a small subset of the original data distribution, ignoring other valid patterns or variations. Repetitive patterns arise when the synthetic data repeats very similar or almost identical to samples across the dataset. To illustrate, if a GAN generates 10,000 synthetic faces, but thousands of them look nearly identical (same pose, expression, background), this shows low variability despite many samples.

Privacy assesses the protection of sensitive information from the original dataset. This encompasses resistance to membership inference attacks, attribute inference attacks, and other privacy breaches while maintaining differential privacy guarantees where applicable.¹⁷

Utility measures the practical usefulness of synthetic data for downstream applications such as machine learning model training, statistical analysis, or decision-making. This is typically evaluated using the Train-on-Synthetic-Test-on-Real (TSTR) paradigm, where models trained on synthetic data are tested on real data to assess performance preservation¹⁸. The goal is not only to make synthetic data look realistic but to ensure that models trained on it behave as if they were trained on the original data.

Scalability evaluates computational efficiency and the ability to generate large datasets. This includes training time, inference speed, memory requirements, and the method's capacity to handle high-dimensional data and large-scale generation.^{19 20}

Compliance measures how well the process of generating, storing, and using synthetic data adheres to applicable legal, ethical, and regulatory frameworks such as GDPR (General Data Protection Regulation), HIPAA (Health Insurance Portability and Accountability Act), and other sector-specific data protection standards. This includes auditability, explainability, and the ability to demonstrate privacy preservation in regulated environments.²¹

¹⁶ <https://www.sciencedirect.com/science/article/pii/S2589004222016030>

¹⁷ <https://pmc.ncbi.nlm.nih.gov/articles/PMC11958975/>

¹⁸ <https://arxiv.org/html/2404.04314v1>

¹⁹ <https://policyreview.info/articles/news/politics-of-synthetic-data-performance-metrics/1761>

²⁰ <https://www.mdpi.com/2079-9292/13/17/3509>

²¹ <https://pmc.ncbi.nlm.nih.gov/articles/PMC11958975/>

Synthetic data generation methods

Synthetic data generation methods can be broadly organised into three main categories based on their underlying algorithmic approaches and mathematical foundations.²²

Generative Methods are often neural network-based approaches that learn to model data distributions and generate new samples. These encompass GANs, VAEs, diffusion models, and autoregressive models like LLMs.²³

Probabilistic Methods use statistical and probabilistic frameworks to model data relationships and dependencies. This category includes Bayesian networks, Gaussian mixture models, and copula-based methods.^{24,25}

Simulation-Based Methods produce synthetic data by explicitly simulating the processes or behaviours underlying real-world systems, instead of learning them solely from observed data.²⁶ This includes physics-based simulations and agent-based models, which simulate interactions between entities to reproduce system-level behaviour. Simulation environments may also support other approaches, such as reinforcement learning, where agents generate sequential data through interaction, though reinforcement learning itself is not a simulation method.

There are also some hybrid approaches that combine multiple methodologies to leverage their strengths. Examples include VAE-GAN combinations, differentially private GANs, and multi-model ensemble approaches.²⁷

A further deep dive into the different synthetic data generation methods can be found in annex I.

²² <https://arxiv.org/html/2302.04062v6>

²³ <https://doi.org/10.3390/electronics13173509>, <https://www.mdpi.com/2079-9292/13/17/3509>

²⁴ <https://pmc.ncbi.nlm.nih.gov/articles/PMC10464868/>

²⁵ <https://arxiv.org/abs/2101.00598>

²⁶ https://royalsociety.org/-/media/policy/projects/privacy-enhancing-technologies/Synthetic_Data_Survey-24.pdf

²⁷ <https://arxiv.org/html/2302.04062v6>

Responsible innovation considerations

While synthetic data can offer significant opportunities for navigating data scarcity and privacy challenges, there are risks with synthetic data that need to be considered when determining the appropriateness of its inclusion in technology projects. This section outlines some of the main identified risk areas that relate to synthetic data in decarbonisation, covering **privacy and re-identification risks**, **safety critical systems risks**, and **bias amplification and fairness risks**, as well as some of the measures being applied to mitigate these risks. However, it should be noted that this is an emerging field where industry is still learning about the nature of risks and the potential of mitigation measures relevant for their synthetic data use cases.²⁸

Privacy and re-identification risks

Synthetic data and generation mechanisms may bring unanticipated privacy challenges, including leaking sensitive information about individuals in the training data and a lack of resilience to membership inference and attribute inference attacks (attempts to determine whether a particular data sample such as an individual, or certain attributes of a data sample, featured in a database).²⁹ With synthetic data drawn from generative models in particular, data representing outliers or members of minority groups have been especially vulnerable to privacy attacks, while those representative of a large majority to the population are often better protected.³⁰ **With decarbonisation and energy sector use cases, this risk area could include synthetic smart meter data revealing sensitive details about household behaviour and occupancy patterns.**

Despite privacy concerns being a primary use case for synthetic data, the risks of data disclosure with synthetic data is impossible to eliminate entirely. Organisations should still apply a risk-based approach for privacy validation based on their specific use case that determines and accepts some level of risk with producing and sharing synthetic data.³¹ Additional consideration may also be needed to consider the acceptable privacy risks for maintaining data utility and fidelity requirements, and where appropriate trade-offs can be made.

Some examples of risk mitigation measures that have been applied in research and industry settings include:

²⁸ <https://www.weforum.org/stories/2025/10/ai-synthetic-data-strong-governance/>

²⁹ https://www.researchgate.net/publication/369700773_A_Unified_Framework_for_Quantifying_Privacy_Risk_in_Synthetic_Data

³⁰ <https://www.usenix.org/system/files/sec22-stadler.pdf>

³¹ <https://arxiv.org/abs/2205.03257>

- Assessing identifiability risks.³²
- Applying privacy evaluation frameworks such as Anonymeter.³³
- Risk assessments that consider different harms arising from different types of synthetic data and differentiated assessments for confidential and personal data.
- Specific validations for the data as well as the data generator.
- Consideration of choices that can increase or mitigate the risks with the data itself, such as how the data is being shared and the barriers to re-identification (e.g. time, cost and available techniques).³⁴

Safety-critical systems risks

Synthetic data use in safety-critical systems can help with addressing challenges such as insufficient real-world data for rare or hazardous scenarios, enabling technological benefits that may not have been possible without synthetic data use.³⁵ There are however significant risks when systems relying on synthetic data are deployed in physical environments and critical infrastructure contexts, where failures could compromise public safety and service stability (known as the 'sim-to-real' gap).³⁶ There have been persistent challenges with ensuring that models and simulations trained or calibrated on synthetic data perform adequately in real world conditions, especially when situations involve edge cases or rare events that are difficult to simulate accurately.³⁷

One form this can take is 'model collapse', where models trained iteratively on synthetic data generated by other AI models progressively lose the ability to represent real world conditions and end up polluting training sets.³⁸ This recursive training can lead to models "forgetting" information about the tails of distribution, data that tends to represent the rare events that are critical for safety and infrastructure resiliency validations.³⁹ **In a decarbonisation context, this could relate to grid management systems failing to recognise unusual consumption patterns or energy forecasting models underperforming during atypical weather conditions.**

Risk mitigation measures for safety-critical synthetic data use, applied from general AI risk mitigation advice, include:

³² <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/about-this-guidance/>

³³ <https://github.com/statice/anonymeter>

³⁴ <https://www.fca.org.uk/publications/research-articles/exploring-synthetic-data-validation-privacy-utility-fidelity>

³⁵ <https://doi.org/10.48550/arXiv.2202.02215>

³⁶ <https://doi.org/10.1016/j.vehcom.2025.100998>

³⁷ <https://doi.org/10.48550/arXiv.1909.11512>

³⁸ <https://arxiv.org/pdf/2307.01850>

³⁹ <https://doi.org/10.1038/s41586-024-07566-y>

- Adopting hybrid validation approaches that combine synthetic data with real-world testing.⁴⁰
- Implementing formal scenario-based testing frameworks aligned with use-case relevant safety standards; maintaining human oversight and fail-safe mechanisms for systems trained on synthetic data.⁴¹
- Ensuring transparency and clear provenance documentation for synthetic data used in safety-critical applications to support audit and accountability requirements.⁴²

Bias amplification and fairness risks

While synthetic data has shown success with addressing bias in datasets,⁴³ there are significant risks of synthetic data generation replicating and amplifying biases present in original data. This has been especially evident when synthetic data is generated using techniques that incorporate differential privacy protections, where the noise-adding mechanisms required for privacy can disproportionately affect underrepresented groups in datasets.⁴⁴

Research examining differential privacy synthetic data generators found that they can either reduce the gaps between majority and minority groups, or increase these gaps, but both types of distributional shifts can lead to disparate impacts on the classification accuracy of synthetic data, with disproportionately greater impacts on the underrepresented subgroups.⁴⁵ **In a decarbonisation context, bias amplification in synthetic data could lead to the underrepresentation of certain households in energy consumption data that have atypical consumption patterns. This may include instances of fuel poverty, medical equipment use or shift workers, and their subsequent exclusion from analyses such as demand forecasting.**

There is also the risk that a synthetic dataset which appears to be fair may not guarantee fair downstream predictions,⁴⁶ or produce fair outcomes in real-world use cases (related to the points covered in the safety-critical systems risks section). In addition to ensuring the data producers have knowledge of the downstream data use and system deployment setting, and that this knowledge is incorporated into fairness criteria and bias definitions for evaluations, approaches to bias risk mitigation could include:

⁴⁰ [https://www.thelancet.com/journals/landig/article/PIIS2589-7500\(25\)00106-2/fulltext](https://www.thelancet.com/journals/landig/article/PIIS2589-7500(25)00106-2/fulltext)

⁴¹ <https://doi.org/10.48550/arXiv.2102.00902>

⁴² <https://www.adalovelaceinstitute.org/blog/synthetic-data-real-harm/>

⁴³ <https://doi.org/10.48550/arXiv.2205.03257>

⁴⁴ <https://doi.org/10.1145/3442188.3445879>

⁴⁵ <https://proceedings.mlr.press/v162/ganev22a.html>

⁴⁶ <https://doi.org/10.48550/arXiv.2304.03722>

- Conducting thorough audits of source data for existing biases before generation.
- Engaging domain experts and affected stakeholders in reviewing synthetic data outputs.
- Implementing ongoing monitoring of downstream model performance across different population segments to detect emergent disparities.

Synthetic data – case studies in action

Below are a range of synthetic data for AI and decarbonisation initiatives sourced through desk research. This list is not designed to be exhaustive but instead indicative of potential future research avenues for the UK AI for decarbonisation ecosystem.

Energy

Faraday⁴⁷

Faraday is a generative AI model developed by the Centre for Net Zero that uses over a billion smart meter readings to produce synthetic household electricity demand profiles conditioned on characteristics such as property type, energy performance rating, low-carbon technology ownership (e.g., EVs, heat pumps), time of year, and tariff type.

Because real smart meter data is highly restricted for privacy reasons, Faraday generates realistic half-hourly consumption data that can be safely shared and used by researchers, policymakers, grid operators and industry to simulate how homes use electricity and to model future energy systems.

The project also supports research into low-carbon grid planning and forms the foundation of the OpenSynth open data community, created with LF Energy, which democratises access to synthetic smart meter datasets globally and has already released millions of synthetic profiles for wider use in energy system analysis.

Faraday has an API for [access](#) to the synthetic dataset and has also enabled [work on evaluation frameworks](#) to benchmark algorithms which generate synthetic smart meter data against fidelity, utility and privacy - to consider whether the data meets privacy requirements whilst still being sufficiently accurate for its intended purpose.

Synthetic-SERL⁴⁸

The Synthetic-SERL project is a six-month EPSRC-funded initiative that aims to unlock access to the valuable SERL energy dataset, which is currently restricted to accredited academic researchers in the UK.

Using Generative Pretrained Transformers (GPTs), the project will produce a synthetic version of the SERL dataset that replicates the statistical patterns and richness of the

⁴⁷ <https://www.centrefornetzero.org/technologies/faraday>

⁴⁸ <https://serl.ac.uk/new-funding-secured-to-develop-synthetic-serl-energy-dataset>

original, including half-hourly electricity and gas usage, building characteristics, occupant profiles, weather, and tariff information while removing all sensitive personal data.

By releasing this synthetic dataset, along with supporting code and tools, the project will enable a much broader group of researchers, policymakers, and industry stakeholders to analyse household energy consumption, test AI models, and explore decarbonisation strategies, without needing access to the gatekept original dataset. This aims to democratise a previously restricted resource, making high-quality energy data widely accessible while preserving privacy.

Home Electricity Data Generator (HEDGE)⁴⁹

HEDGE (Home Electricity Data Generator) is an open-access tool that generates realistic synthetic residential energy data – specifically daily profiles for household electricity demand, rooftop solar (PV) generation, and electric vehicle consumption and availability, based on real UK datasets to fill gaps where long, consistent sequences of real home energy data are scarce or incomplete.

To produce this synthetic data, HEDGE preprocesses raw datasets to fill missing values and cluster households by usage behaviours, then trains generative adversarial networks (GANs) to learn realistic patterns within these clusters, and finally uses a Markov chain mechanism combined with the GAN outputs to generate multiday profiles that reflect real behavioural and physical patterns. The tool addresses a major barrier for research on distributed energy resources and data-driven energy system methods – lack of usable data – by providing consistent, realistic synthetic sequences that can support forecasting, control, and coordination studies in residential energy research.

HEDGE is hosted on [GitHub](#), and offers a way to randomly generate realistic home photovoltaic (PV) generation, electricity consumption, and electric vehicle consumption and availability daily profiles, based on UK historical datasets. It can for example generate data that can be used to train neural networks.

Brunel University - Synthetic Electricity Consumption Data Generation using Tabular Generative Adversarial Networks⁵⁰

This paper proposes using Tabular Generative Adversarial Networks (Tabular GANs) to generate realistic synthetic electricity consumption data for residential users, addressing

⁴⁹ <https://arxiv.org/abs/2310.01661>

⁵⁰ <https://bura.brunel.ac.uk/bitstream/2438/29977/1/FullText.pdf>

challenges around privacy, data scarcity, and incomplete real datasets that limit smart grid research and energy system modelling.

The authors trained a Tabular GAN on cleaned and pre-processed smart meter and weather data from the Low Carbon London dataset to produce synthetic half-hourly load profiles that closely match the statistical properties (mean, variance, distribution) and temporal patterns of the original data. Results show that the synthetic data preserves key consumption behaviours and correlations with features like time-of-day, temperature, and calendar effects, demonstrating that Tabular GANs can produce high-quality synthetic consumption profiles that are suitable for simulating demand and supporting AI-enabled analysis in smart energy systems.

Agriculture

AgriSynth UK⁵¹

AgriSynth is an AI project that uses synthetic image generation to create high-quality datasets for training machine learning models. It focuses primarily on detecting blackgrass in wheat crops, addressing the common problem of data scarcity in agricultural AI development. Rather than relying on costly and logistically complex fieldwork, the project generates diverse, realistic images that cover a wide range of agricultural scenarios, conditions, and crop stages.

A key advantage of synthetic data is that every image comes with pixel-perfect annotations, something that is extremely difficult and time-consuming to achieve with real-world imagery. This precision improves the effectiveness of AI models and opens up opportunities for startups and research institutions that may lack the resources for traditional data collection. The approach also allows models to be refined in a controlled environment before being deployed in real agricultural settings.

The project is an 18-month Innovate UK SMART initiative led by AgriSynth UK, in partnership with Real-Time Tech and the UK Agri-Tech Centre. Beyond solving immediate challenges like blackgrass detection, it aims to future-proof agri-tech innovation by supporting more sustainable farming practices including reduced pesticide use through early disease detection and better resource optimisation through precision agriculture, contributing to food security and climate goals.

⁵¹ <https://agrisynth.io/>

A Synthetic Data Generation Pipeline for Improving the Segmentation of Roots in Micro-CT Images of Soil.⁵²

An important step in harnessing visual AI in agriculture requires training a computer vision model to be able to distinguish between different subjects within an image, e – such as identifying roots against soil in scans. This identification allows downstream measurement or calculations that enable various elements of precision agriculture, an important part of sustainable agriculture.

Training AI models to identify plant roots in micro-CT scans requires large amounts of accurately labelled images, but creating these manually is time-consuming and can take hours or days to annotate a single high-resolution scan. This research project explored synthetic data to generate realistic-looking images of tomato roots in soil using a free 3D modelling tool called Blender. Image labels denoting roots and soil were created automatically during the generation process.

The synthetic images were combined with small amounts of real scan data to train a segmentation model that could identify and outline roots within a soil image. The researchers found that when real data is scarce, adding synthetic images meaningfully improved the model's accuracy. The quality of the synthetic textures also matters and images with grey-value distributions that more closely matched real scans produced better results. The pipeline is also designed to be modular and adaptable, meaning it could be applied to different root types, soil compositions, or even other biological structures beyond this specific experiment.

This project found that synthetic data is most valuable when real annotated data is limited as it can fill the gap without requiring costly fieldwork or manual labelling. However, having at least some real data in the training set remains important, as purely synthetic training still underperforms. The overall segmentation accuracy was modest, partly because roots can look very similar to pores or debris in CT scans, and fine root tips were easily missed. The authors see this project as a proof of concept, with future work focused on refining the texturing process, incorporating more environmentally realistic root models, and exploring 3D segmentation approaches.

⁵² <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70222>

Built Environment

Generating Synthetic Energy Usage Data to Enable Machine Learning for Sustainable Accommodation⁵³

This paper explores the applicability of synthetic data in training ML models for identifying correlations between energy usage in apartments and variables such as seasonality, apartment location, size, efficiency and occupant characteristics. These insights can enable apartments to become more energy efficient and allow occupants to save energy.

ML models for understanding and improving energy use in apartments requires sufficient data, but in many real-world settings, adequately long histories of energy usage are not available. To address this, the authors created a digital twin-based agent-driven synthetic data generator that uses expert knowledge of how building characteristics (like size, location, efficiency and occupant behaviour) influence energy consumption to generate unlimited realistic synthetic energy usage data, even when no real measurements exist.

paper outlines a case study involving corporate and luxury accommodation (TheSqua.re), and demonstrates that synthetic data can be produced for different scenarios. It also highlights how synthetic data can be and is suitable for training machine-learning models to uncover correlations between features and energy demand, thus enabling AI tools to be developed sooner and more effectively for sustainable building operations.

Synthetic Population Catalyst⁵⁴

Developing population-level models for transport, health and urban planning requires access to detailed data about how individuals live, work and move around. In practice, this data is fragmented across many sources like census records, health surveys, employment registers, travel diaries, with each maintained separately, in different formats, and at different geographic scales. This often means combining the different datasets can often be time-consuming and technically demanding.

The Synthetic Population Catalyst (SPC) is a Turing-led initiative funded by UKRI and EPSRC that fuses disparate datasets into a single, pre-processed synthetic population file covering any area in Great Britain. Each synthetic individual is assigned attributes drawn statistically from real sources such as age, sex, household type, car ownership, employment, income, health conditions, and a daily time-use diary, without corresponding to any real person. The underlying data comes from the 2011 census, the UK Time Use

⁵³ https://discovery.ucl.ac.uk/id/eprint/10141464/1/ICECCME_v10.pdf

⁵⁴ <https://alan-turing-institute.github.io/uatk-spc/>

Survey, national health surveys, and business employment registers, with workplace locations and commuting flows modelled using established spatial interaction methods.

For decarbonisation, the SPC provides important population data for transport and built environment research, modelling travel behaviour at the individual level including how different demographic groups might respond to changes in public transport provision, active travel infrastructure, or land use. The inclusion of car ownership, income and employment data makes it particularly useful for assessing who is realistically able to shift away from private car use, which is central to transport decarbonisation. It can also support urban heat and climate resilience planning by identifying where vulnerable populations are concentrated, and serves as a population input layer for urban digital twins.

Transport

SynSafeAV⁵⁵

The SynSafeAV project at Coventry University investigates the systemic risks of using AI-generated synthetic data in autonomous vehicle (AV) development, focusing on how biases, inaccuracies or unrealistic patterns in synthetic datasets can compromise AV performance and lead to broader impacts on traffic flow, safety outcomes and social equity.

The research uses machine learning, transport simulations and socio-demographic data to identify, categorise, and quantify technical and social hazards. It aims to produce a risk assessment framework with standardised evaluation methods and policy recommendations to improve safety, governance and equitable deployment of AV systems.

Although the project's primary aim is not directly about decarbonisation, its outcomes will support the safe and trustworthy adoption of autonomous vehicles and smart mobility technologies, which in turn can enable more efficient transport systems, higher uptake of shared and automated mobility services, and reduced emissions through improved traffic flow, fewer collisions, and better integration with electrified transport – all of which contribute to broader decarbonisation goals.

⁵⁵ <https://www.coventry.ac.uk/research/research-directories/projects/2025/synsafeav/>

Manufacturing

Generating synthetic data to train accurate AI object detection⁵⁶

This is a case study from the University of Sheffield's Advanced Manufacturing Research Centre which built an AI model that could automatically generate labelled synthetic images to train AI object detection models. Rather than photographing and manually labelling hundreds of real objects, the tool takes existing CAD (computer-aided design) data and uses it to render thousands of realistic images automatically, complete with pixel-perfect labels. They demonstrated it by training an AI to recognise surgical tools, generating 600 labelled images in under an hour; a task that would traditionally take a human engineer five or more hours.

The AI model trained entirely on synthetic renders was able to successfully detect real-world objects it had never seen before, performing comparably to models trained on real photographs. Occasionally the system confused similar-looking tools like scissors and forceps, but the results were strong enough for a proof of concept. The team identified future improvements including adding simulated wear and defects to the CAD models, which could make the tool useful for detecting faults or damage on industrial parts.

The ability to train AI models cheaply and quickly using only CAD data and without physical prototypes or extensive testing, has potential in industrial decarbonisation efforts. For example, the tool could be used to train defect detection systems for components in renewable energy infrastructure (wind turbines, solar panels, EV batteries), where catching faults early reduces waste and improves efficiency. More broadly, reducing the cost and time of developing AI-powered inspection and monitoring systems lowers the barrier to adopting smart manufacturing processes, which are central to making industrial production cleaner and more resource-efficient.

⁵⁶ https://www.amrc.co.uk/files/document/466/1647421007_GeneratingSyntheticData.pdf

Conclusions and recommendations

This review has outlined synthetic data generation methods, responsible AI considerations, and illustrated where the focus lies for synthetic data and AI for decarbonisation in the UK landscape.

Synthetic data generation is a potential enabler of privacy-preserving analytics and AI model training. Different synthetic data generation methods offer a balance between realism, scalability, and privacy, and selecting the appropriate approach depends on the specific use case, data sensitivity, and desired trade-offs among correctness, diversity, privacy, utility, fidelity, scalability, and compliance.

Ultimately, combining these methods or adopting hybrid architectures often yields the best results in balancing realism with responsible AI principles.

Synthetic does introduce risks that require careful management. Privacy and re-identification risks remain difficult to eliminate, particularly for minority or outlier data points. Safety-critical applications face the sim-to-real gap, where models trained on synthetic scenarios fail to perform reliably in unpredictable real-world conditions, and fairness risks arise when generation processes replicate or amplify bias.

Addressing these risks require governance and technical measures.⁵⁷ This includes imbedding rigorous validation and transparency into workflows, maintaining clear provenance for synthetic datasets, and applying scenario-based stress testing aligned with relevant safety standards to reduce performance gaps. Continuous monitoring for bias and engaging diverse stakeholders in evaluation can help to ensure synthetic data supports innovation without compromising trust and safety.

In a decarbonisation context, these practices are critical for applications such as grid management and energy demand forecasting, where failures caused by unrealistic synthetic scenarios or amplified biases could undermine system resilience and equitable access to energy.

The landscape for synthetic data and decarbonisation applications is not well defined in the UK and characterised by isolated projects across diverse objectives. There could be an issue of language in that 'synthetic data' as a specific term is being used interchangeably in projects involving 'simulation' or 'digital twinning'. These projects could be incorporating elements of synthetic data generation but were difficult to identify within the

⁵⁷ https://reports.weforum.org/docs/WEF_Synthetic_Data_2025.pdf

methodological constraints of this review. However, the lack of shared terminology in and of itself indicates a fragmented landscape.

The progression of synthetic data adoption could benefit from a more coherent strategy or programme that demonstrates and advances the practical application of synthetic data.

1. Join up efforts on synthetic energy usage datasets

Multiple UK based projects have explored creating AI and ML ready synthetic datasets for energy usage of residential and commercial buildings, but there is little evidence of a joined-up approach between these projects.

ADViCE and other stakeholders such as [OpenSynth](#) should work together to align these efforts and ensure research does not happen in siloes, particularly when so many of these projects exhibit similar goals and objectives.

2. Image-generation shows promise across multiple sectors and applications

A key, cross sectoral use case identified in this report relates to the synthetic generation of images and by extension, video data that can be used in a range of applications e.g., training autonomous vehicles, crop disease detection, or identifying manufacturing faults. These applications are important in improving overall efficiency of systems, with downstream consequences for net zero.

A consolidated programme looking at cross sectoral applications of synthetic image or video-generated datasets to enable the following:

- Provide a testing ground to assess different methods for generating image- or video-based synthetic datasets against the key criteria mentioned above.
- Develop assurance mechanisms tailored specifically for image- or video-based synthetic datasets.
- Provide best practice guidance for the creation of image- or video-based synthetic datasets
- Accelerate reuse of methods and tools that have been successful in one context into other sectors.

By doing so this programme could unlock cross-sectoral scaling pathways for innovators exploring synthetically trained computer vision models to push beyond PoCs into commercialisation.

3. Ecosystem mobilisation and understanding

ADViCE has identified [100+ challenges](#) whereby AI can play a role in decarbonising the UK economy. Through ADViCE and other research projects, these challenges can be analysed to find areas that exhibit data barriers that align with the key use cases of synthetic data, namely privacy concerns, data scarcity and reducing barriers to entry.

This analysis could identify targets for synthetic data innovation activity and mobilise communities towards collective action.

As ADViCE engages its ecosystem around these challenges the programme will actively investigate the potential for synthetic data alongside other AI related technologies in driving towards net zero.

4. Monitor for learnings from synthetic data projects in other sectors and international exemplars

There are synthetic data initiatives in the UK focused on other sectors such as the MHRA and FCA synthetic data trials which, whilst not decarbonisation specific still offer learnings for synthetic data best practice.^{58,59,60}

International projects identified that are specific to decarbonisation have focused on areas such as earth observation,⁶¹ energy grid derived synthetic datasets,⁶² and extreme wind events.⁶³

Whilst these projects were beyond the scope of this initial review, they represent potential exemplars to feed into future synthetic data research.

⁵⁸ <https://www.phgfoundation.org/wp-content/uploads/2025/07/Synthetic-data-for-development-of-AI-as-a-medical-device.pdf>

⁵⁹ <https://www.gov.uk/government/news/side-effects-from-drug-interactions-to-be-predicted-by-ai-before-reaching-patients>

⁶⁰ <https://www.fca.org.uk/publications/corporate-documents/report-using-synthetic-data-financial-services>

⁶¹ <https://arxiv.org/pdf/2112.02829>

⁶² <https://arxiv.org/pdf/2110.06324>

⁶³ <https://www.sciencedirect.com/science/article/abs/pii/S0960148123016841>

Annex I

Overview of State-of-the-Art Techniques

Generative Adversarial Networks (GANs)

GANs represent one of the most widely adopted approaches for synthetic data generation, employing an adversarial training framework where a generator network creates synthetic data while a discriminator network attempts to distinguish between real and synthetic samples.⁶⁴ Over many iterations, the generator improves so much that the discriminator network can no longer tell the difference.

GANs are highly proficient in generating high-quality synthetic data, particularly for images, tabular data, time series where realism matters (e.g., energy usage patterns, faces, product photos). However, they face challenges with being hard to train, can produce repetitive samples, and might leak real data if overfit, that is, the generator of the GAN model fails to generalise due to overfitting leading to a lack of diversity and sometimes duplicating or making a direct copy of the actual data which compromises the privacy of the original data subjects by revealing their specific information.⁶⁵ This is a significant concern when GANs are used to create synthetic data for cases where the original data holders are unwilling to share their actual samples, such as in medicine or finance.

GANs have shown promising results in generating synthetic smart meter data and energy consumption patterns for grid optimisation and demand forecasting applications.^{66,67} Their ability to capture complex temporal patterns makes them suitable for modelling energy system dynamics.

Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs) provide a probabilistic approach to synthetic data generation, learning a latent distribution that captures the key underlying structure of the real dataset.⁶⁸ In contrast to deterministic autoencoders,⁶⁹ VAEs learn a distribution over

⁶⁴ <https://www.mdpi.com/2079-9292/13/17/3509>

⁶⁵ <https://datasciencecampus.ons.gov.uk/projects/generative-adversarial-networks-gans-for-synthetic-dataset-generation-with-binary-classes/>

⁶⁶ <https://www.sciencedirect.com/science/article/pii/S2666546822000209>

⁶⁷ <https://bura.brunel.ac.uk/bitstream/2438/29977/1/FullText.pdf>

⁶⁸ https://link.springer.com/chapter/10.1007/978-3-031-24628-9_16

⁶⁹ <https://arxiv.org/abs/1312.6114>

latent variables so that new samples can be generated by sampling from this latent space and decoding back into the data space.

VAEs work well with structured numeric and time-series data, where controlled variation and a smooth latent representation are important for capturing overall patterns and correlations. However, they tend to produce smoother or blurred outputs compared with adversarial approaches because the objective emphasises learning a broad latent distribution rather than sharp pointwise detail.⁷⁰

VAEs have been applied in several domains involving complex time-series and structured data. For example, research on synthetic energy time series generation identifies VAEs as one of the methods used to model and produce energy consumption patterns that resemble real data distributions. Additionally, hybrid VAE-based techniques (e.g., VAE-GANs) have been proposed to generate synthetic electricity load profiles for smart home energy management systems and demonstrate the capacity of VAEs as part of effective synthetic generation frameworks for energy applications.^{71,72,73}

Diffusion Models

Diffusion models have recently emerged as state-of-the-art generative models, achieving leading performance in image synthesis and showing increasing promise for other data modalities such as tabular and time-series data. Diffusion models operate through a forward noising process and a learned reverse denoising process, where noise is gradually added to training data and a neural network is trained to reverse this process. At generation time, the model begins with random noise and iteratively denoises it to produce realistic samples.^{74,75}

Recent advances include Denoising Diffusion Probabilistic Models (DDPM) and score-based generative models, which provide a strong theoretical foundation and stable training dynamics compared with earlier generative modelling approaches.^{76,77} These methods have demonstrated excellent sample quality and robustness, avoiding many of the training instabilities associated with generative adversarial networks while achieving

⁷⁰ <https://www.mdpi.com/2079-9292/13/17/3509>

⁷¹ <https://arxiv.org/pdf/2305.08885>

⁷² <https://www.sciencedirect.com/science/article/pii/S1364032124005689>

⁷³ <https://arxiv.org/abs/2411.03936>

⁷⁴ <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>

⁷⁵ <https://arxiv.org/abs/2102.09672>

⁷⁶ <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>

⁷⁷ <https://arxiv.org/abs/2011.13456>

comparable or superior results in several generative modelling tasks.^{78,79} However, diffusion models typically require many sequential denoising steps during sampling, which can make generation slower compared with alternative approaches such as GANs.⁸⁰

In energy systems research, diffusion-based approaches are increasingly being explored for generating synthetic weather patterns, renewable energy generation scenarios, and load profiles, where maintaining realistic temporal dynamics is essential for modelling and forecasting applications.⁸¹

Large Language Models (LLMs)

Large Language Models (LLMs) have significantly expanded the capabilities of synthetic data generation, particularly for text and structured tabular data, by leveraging large-scale pre-training and few-shot learning capabilities.^{82,83} These models can generate synthetic datasets by conditioning outputs on structured prompts or examples, enabling them to maintain semantic coherence and adapt to a wide range of domains and data formats.^{84,85}

Recent work demonstrates that LLMs can produce synthetic tabular datasets through prompt-based generation, where structured prompts guide the model to create records that reflect the relationships and distributions observed in the original data.⁸⁶ LLM-based synthetic data generation often relies on techniques such as prompt engineering, few-shot learning, retrieval-augmented generation (RAG), and iterative refinement, which help improve the realism and domain specificity of generated samples.⁸⁷

Despite these capabilities, LLM-generated synthetic data presents several challenges. Studies highlight limitations in preserving statistical fidelity, maintaining privacy guarantees, and avoiding hallucinations or unrealistic patterns introduced during

⁷⁸ <https://arxiv.org/pdf/2105.05233>

⁷⁹ <https://arxiv.org/abs/1907.05600>

⁸⁰ <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>

⁸¹ <https://arxiv.org/html/2507.14507v1>

⁸² <https://arxiv.org/abs/2005.14165>

⁸³ <http://arxiv.org/abs/2212.10560>

⁸⁴ <https://arxiv.org/pdf/2410.21717>

⁸⁵ <https://dl.acm.org/doi/10.1145/3756681.3756985>

⁸⁶ <https://arxiv.org/pdf/2410.21717>

⁸⁷ <https://arxiv.org/abs/2005.11401>

generation.^{88,89,90,91} These issues can reduce the reliability of synthetic datasets if not carefully evaluated and validated.

Finite Mixture Models (FMMs)

Finite mixture models provide a classical probabilistic approach to synthetic data generation by modelling data as combinations of multiple probability distributions.^{92,93} In practice, the most used form is the Gaussian Mixture Model (GMM), where the data distribution is approximated as a weighted combination of Gaussian components.⁹⁴ These models assume that complex datasets can be represented as several underlying sub-populations, each described by a probability distribution.⁹⁵ For example, if customer income data contains low-, middle-, and high-income groups, a mixture model can learn these underlying groups and generate new synthetic samples from the learned distributions.

Although Gaussian components are widely used due to their mathematical convenience and efficient estimation, mixture models can also employ other distributions depending on the data type, such as Poisson distributions for count data or Bernoulli distributions for binary variables.⁹⁶

Mixture-based approaches offer computational efficiency, interpretability, and strong theoretical foundations. They are particularly effective when data exhibits multi-modal distributions and when explicit control over generation parameters is desired. However, they tend to struggle with very high-dimensional or highly complex data compared with modern deep generative models.^{97,98}

⁸⁸ <https://arxiv.org/abs/2306.15895>

⁸⁹ <https://arxiv.org/abs/2412.02980>

⁹⁰ <https://arxiv.org/abs/2412.03679>

⁹¹ <https://aclanthology.org/2024.findings-acl.658.pdf>

⁹² <https://academic.oup.com/jrsssb/article/39/1/1/7027539>

⁹³ <https://sites.stat.washington.edu/raftery/Research/PDF/fraley2002.pdf>

⁹⁴ <https://academic.oup.com/jrsssb/article/39/1/1/7027539>

⁹⁵ <https://sites.stat.washington.edu/raftery/Research/PDF/fraley2002.pdf>

⁹⁶ <https://www.annualreviews.org/content/journals/10.1146/annurev-statistics-031017-100325>

⁹⁷ <https://www.sciencedirect.com/science/article/pii/S095741741101709X>

⁹⁸ <https://arxiv.org/pdf/2510.25001>

Copulas

Copulas provide a flexible framework for modelling dependencies between multiple variables while allowing the use of marginal distributions.^{99,100} In this framework, the marginal distributions are estimated separately, and the copula captures the joint dependency structure that links them. This allows the generation of new synthetic samples that preserve realistic inter-variable relationships. For example, in finance, copula-based generators can simulate asset returns or credit risk profiles that maintain correlations observed in real data such as higher interest rates being associated with increased loan defaults.^{101,102,103}

Modern copula-based approaches extend simple parametric copulas to include vine copulas and copula flows, which can handle high-dimensional datasets with complex dependency patterns.^{104,105,106} By modelling dependencies rather than individual records directly, copula-based synthetic data generation also offers privacy advantages, making it suitable for sensitive applications.¹⁰⁷

Copula-based methods are particularly well-suited for financial, risk, and other multivariate applications where interdependent variables are central. However, they require careful parameter tuning and can become computationally intensive for very high-dimensional datasets.^{108,109}

Bayesian Networks

Bayesian networks model probabilistic relationships between variables using directed acyclic graphs (DAGs), enabling transparent and interpretable synthetic data generation.^{110,111} These networks encode conditional dependencies between variables in a

⁹⁹

https://www.researchgate.net/publication/2397178_Modelling_Dependence_With_Copulas_and_Applications_to_Risk_Management

¹⁰⁰ <https://www.sciencedirect.com/science/article/abs/pii/S0167668707000108>

¹⁰¹ <https://www.sciencedirect.com/science/article/pii/S0047259X12000826>

¹⁰² https://link.springer.com/chapter/10.1007/978-1-4471-7503-2_46

¹⁰³ https://www.researchgate.net/publication/220504759_Modeling_Multivariate_Count_Data_Using_Copulas

¹⁰⁴ <https://arxiv.org/pdf/2501.13304v1>

¹⁰⁵ https://www.researchgate.net/publication/228435306_Dependence_modeling_with_C-and-D-vine_copulas_The_R-package_CDVine

¹⁰⁶ <https://arxiv.org/abs/2101.00598>

¹⁰⁷ <https://petsymposium.org/popets/2021/popets-2021-0040.pdf>

¹⁰⁸ <https://www.sciencedirect.com/science/article/pii/S2090447924004350>

¹⁰⁹ <https://arxiv.org/abs/2409.18611>

¹¹⁰ <https://www.sciencedirect.com/science/article/pii/B9780080514895500084>

¹¹¹ <https://link.springer.com/article/10.1023/A:1007465528199>

graph structure, effectively describing how one variable can influence another. Once the network is learned, it can generate synthetic data by sampling from the joint distribution implied by the DAG. For example, in healthcare, Bayesian networks have been used to simulate disease risk outcomes based on demographic and clinical factors.

Modern Bayesian network approaches leverage structure learning algorithms to automatically discover relationships and conditional dependencies from observed data, improving scalability and reducing reliance on expert-defined structures.^{112,113} Bayesian networks also offer strong interpretability and privacy advantages, as the model captures dependencies statistically rather than storing raw individual records, making them suitable for regulated environments.^{114,115}

Bayesian networks are particularly effective in domains where causal reasoning, interpretability, and privacy are important, such as healthcare and finance.^{116,117,118} However, they can be computationally intensive on large datasets, and defining accurate structures for complex systems often requires domain expertise.

Reinforcement Learning & Simulation

Simulation-based methods generate synthetic data by explicitly modelling the processes or behaviours underlying real-world systems.^{119,120} This includes physics-based simulations, agent-based models, or other environment models, which can produce realistic data without relying on existing datasets. For example, a traffic simulation can model driver behaviour and generate synthetic traffic patterns, while an energy system simulation can produce consumption or generation profiles.

Reinforcement Learning (RL), in contrast, is a framework for training agents through trial-and-error interaction with an environment.¹²¹ While RL itself does not inherently generate synthetic data, it can leverage simulations to learn policies, which in turn may be used to produce sequences of synthetic data (e.g., trajectories in robotics or process control

¹¹² <https://doi.org/10.1023/A:1022623210503>

¹¹³ <https://arxiv.org/abs/2002.00269>

¹¹⁴ <https://www.medrxiv.org/content/10.1101/2020.07.15.20154286v1.full.pdf>

¹¹⁵ <https://bura.brunel.ac.uk/bitstream/2438/22776/1/FullText.pdf>

¹¹⁶ <https://arxiv.org/abs/2211.15258>

¹¹⁷ <https://www.scb.se/contentassets/ca21efb41fee47d293bbee5bf7be7fb3/using-bayesian-networks-to-create-synthetic-data.pdf>

¹¹⁸ <https://arxiv.org/html/2402.17915v1>

¹¹⁹ https://doi.org/10.1007/978-3-031-70415-4_7

¹²⁰ <https://arxiv.org/pdf/2508.13989>

¹²¹ <https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

tasks).¹²² Modern approaches often integrate adaptive, goal-driven strategies in which the agent optimises for specific objectives, adjusts generation strategies based on feedback, and improves data realism over time which can be thought to be conceptually similar to emerging Agentic AI ideas¹²³.

Stepwise RL approaches break complex generation tasks into sub-trajectories, enabling more controlled learning and sampling. These methods are particularly effective at producing sequential or dynamic data requiring temporal consistency or causal relationships, such as energy systems¹²⁴, robotics, games, or process control¹²⁵. However, simulations can be computationally expensive, and RL-based synthetic data generation often relies on high-fidelity environments, which may be costly or complex to construct. Careful design of simulation environments and learning strategies is essential to generate high-quality synthetic data efficiently.

Summary of Criteria for Different Synthetic Data Generation Methods

Below two tables show a summary of criteria in a general sense for different methods that has been found through research and may vary based on use case, approach and data.

Method	Correctness	Diversity	Privacy	Utility	Fidelity	Scalability	Compliance
GANs	Usually good for images or structured data but may miss rare patterns.	Can sometimes repeat patterns also known as mode collapse.	May leak information from training data if not carefully handled.	Good for model training if trained well.	High visual realism for images.	Can generate large datasets, but training is computationally heavy.	Needs extra measures to ensure privacy laws are met.
VAEs	Captures overall patterns well but may smooth out extreme values.	Usually better than GANs at avoiding repeated samples.	Fairly good since it generates from compressed summaries.	Good for model training, works well for continuous data.	Moderate realism but may blur fine details.	Efficient and fast for large datasets.	Reasonable if data is pre - processed to remove sensitive Information.

¹²² <https://arxiv.org/html/2601.22607v1>

¹²³ https://www.dfki.de/fileadmin/user_upload/import/13272_SynTiSeD_paper.pdf

¹²⁴ <https://ieeexplore.ieee.org/abstract/document/10123429>

¹²⁵ <https://arxiv.org/html/2601.22607v1>

Diffusion Models	Very accurate in capturing data patterns.	High diversity and less risk of repetition.	Similar to VAEs and generally safe.	Excellent for generating training data.	Very high quality, often better than GANs.	Slower to generate than VAEs or GANs.	Can be privacy compliant with proper controls.
LLMs	Great for text or structured data; sometimes hallucinate unrealistic info.	High; can generate many variations.	Risk of leaking training data if not careful.	Excellent for NLP and tabular data modelling.	Captures style and meaning well.	Very fast once pre-trained.	Needs prompt design and privacy filtering for regulations.
GMMs	Good for continuous data but may struggle with very complex patterns.	Moderate depends on the number of curves used.	Safe, because it doesn't memorise individual records.	Useful for statistics and basic modelling.	Good for simple data distributions.	Very fast and lightweight.	Easy to make it compliant.
Copulas	Excellent at capturing relationships between variables.	High if designed well.	Good as there is no direct copying of real records.	Very useful for complex tabular data.	Captures statistical structure well.	Can be computationally heavy for many variables.	Naturally compliant if handled properly.
Bayesian Networks	Very accurate for structured relationships.	Moderate, depending on model complexity.	Strong, does not memorise records.	Excellent for predictive modelling and decision support.	High for relational structure, not visual realism.	Can slow down with many variables.	Excellent for regulated domains due to transparency.

Reinforcement Learning / Simulation	Good for sequential and causal data.	High if scenarios are varied.	Safe as it does not rely on real data.	Excellent for time-series and scenario modelling.	Can mimic real processes well.	Depends on simulation complexity.	Compliant as no real data is directly used.
Agentic AI	Can be highly accurate if trained well.	Very high as it can adapt to cover gaps.	Safe due to independent generation.	Promising for adaptive, context-aware applications.	Can capture realistic dynamics and behaviours.	Emerging field and efficiency may vary.	Potentially compliant with proper design.

In conclusion generative approaches such as GANs, VAEs, and diffusion models excel in producing highly realistic data but require careful handling to ensure stability and prevent privacy leakage. Probabilistic and statistical models like GMMs, copulas, and Bayesian networks provide interpretability, strong privacy guarantees, and are well suited for structured or tabular data. LLMs extend synthetic generation into text-rich and mixed data domains, while simulation-based and agentic AI methods enable dynamic, context-aware data creation for complex systems.